Week of 31st

Monday, 30 October 2023 4:51 PM

Realistic, & practical flow :

Turbulent flow: can use the same equations (microscopic balances)

↳ However, flow is no longer unidirectional & SS } time dependent and 3D → 3D PDE

CFD

$v_\theta, v_r \neq 0$

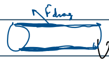
↓

Cannot solve for velocity profile!

↓

v_z $\vec{z}$

v_r , v_θ



Use experiments to fill in the gaps! → {Empirical correlations} → darcy friction

Pressure drop vs. flow rate relationships
(F_{drag}) vs. (Q)

Fluid force on a surface S : $\underline{F} = \iint_S [\hat{n} \cdot \underline{\Pi}]\Big|_{@surface} dS$

This force is being exerted by the fluid on the surface

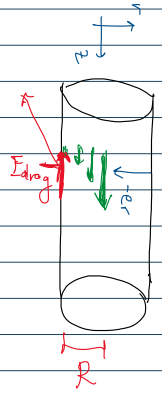
↓

This is the cause of pressure drop in piple flow!!

$\underline{F} \Rightarrow$ total force

$\underline{F}_{drag} = \underline{F}_z$ (i.e. z^{th} component of force)

$$\therefore \underline{F}_{drag} = \int_0^L \int_0^{2\pi} \hat{e}_z \cdot \left(-\hat{e}_r \cdot \underline{\Pi}\Big|_R\right) R \, d\theta \, dz$$



↓ non-dimensionalize

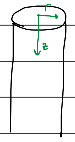
f

$$\boxed{\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu \left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z} \right)\Bigg|_R R \, d\theta \, dz}$$

Fluid drag in turbulent flow on a piper surface of unit normal $-\hat{e}_r$

Practice Questions

Laminar flow in a pipe: (Unidirectional, SS) } Continuity + Navier → BC → v profile



$$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z} ; \quad \underline{g} = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}_{r\theta z} \qquad p = p(z)$$

↓

+ Pressure profile: $p(z) = -\left(\frac{P_0 - P_L}{L}\right) z + P_0$

+ Velocity profile: $v_z(r) = \frac{(P_0 - P_L + \rho g L) R^2}{4\mu L} \left[1 - \left(\frac{r}{R}\right)^2 \right]$

$$\underline{\Pi}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$\Rightarrow Q = \langle v \rangle \pi R^2 = \frac{\pi (P_0 - P_L) R^4}{8\mu L}$$

$$= -p\underline{\underline{I}} + \begin{pmatrix} 0 & 0 & \mu\frac{\partial v_z}{\partial r} \end{pmatrix}$$

$$\underline{\Pi}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$= -p\underline{\underline{I}} + \begin{pmatrix} 0 & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \mu\frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix}_{r\theta z}$$

$$= \begin{pmatrix} -p(z) & 0 & (p_L-p_0)\frac{r}{2L} \\ 0 & -p(z) & 0 \\ (p_L-p_0)\frac{r}{2L} & 0 & -p(z) \end{pmatrix}$$

$$\rightarrow Q = \langle v\rangle \pi R^2 = \frac{\pi(p_0-p_L)R^4}{8\mu L}$$

$$\langle v\rangle = \frac{(p_0-p_L)R^2}{8\mu L}$$

→ Neglect gravity:

$$v_z(r) = \frac{(p_0-p_L+\overset{0}{\cancel{\rho g}}L)R^2}{4\mu L}\left[1-\left(\frac{r}{R}\right)^2\right]$$

$$= \frac{(p_0-p_L)R^2}{4\mu L}\cdot\frac{2}{2}\left[1-\left(\frac{r}{R}\right)^2\right]$$

$$v_z(r) = 2\langle v\rangle\left[1-\left(\frac{r}{R}\right)^2\right]$$

$$\frac{\partial v_z}{\partial r} = \frac{-4\langle v\rangle r}{R^2}$$

$$F_{drag} = \int_0^L\int_0^{2\pi} -\mu\left(\frac{\partial v_z}{\partial r} + \overset{0}{\cancel{\frac{\partial v_r}{\partial z}}}\right)\Bigg|_R R\,d\theta\,dz$$

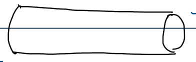
$$= \int_0^L\int_0^{2\pi} -\mu\left(\frac{-4\langle v\rangle r}{R^2}\right)\Bigg|_R R\,d\theta\,dz$$

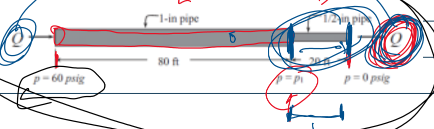
$$= (4\mu\langle v\rangle)(2\pi\cdot L)$$

$$= \cancel{4}\cancel{\mu}\,\cancel{2}\pi\cancel{L}\cdot\left(\frac{(p_0-p_L)R^2}{\underset{2}{\cancel{8\mu L}}}\right)$$

$$\boxed{F_{drag} = \pi R^2(p_0-p_L)}$$

Turbulent flow in smooth pipes:





Find an Q

ΔP₁ → f → New Q

Reynolds

Prandtl correlation 7.156

$$\frac{1}{\sqrt{f}} = 4\log\left[Re\sqrt{f}\right] - 0.4$$ (Smooth pipes)

friction

$$Re = \frac{\rho\langle v\rangle D}{\mu}$$

f

⟨v⟩ ??

$$\frac{1}{\sqrt{f}} = \underbrace{4\log\left[Re\sqrt{f}\right] - 0.4}_{g(f)}$$

Algorithm 1:

$Q = \langle v\rangle \cdot A$

$$f = \frac{\Delta p D}{2L\rho\langle v\rangle^2}$$ (Pipe friction factor from experiments)

drag friction factor

f in Pradtl

1) Guess Q_{guess} (will give ⟨v⟩)
2) Find Re
3) Find f_1 — prandtl correlation
4) Find Δp_1
5) Calculate Δp_2 = known value − Δp_1 (60 psig)
6) Solve for "LHS" (given Δp_2) → gives f_2 → ⟨v⟩ (using friction factor)

friction factor

$$\Rightarrow 4\log\left[\frac{\rho\cancel{\langle v\rangle}D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho\cancel{\langle v\rangle^2}}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4\log\left[\frac{\rho D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4\log\left[\sqrt{\frac{\rho\Delta p D^3}{2L\mu^2}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

→ guess f

→ g(f)

$$\rightarrow f_{new} = \frac{1}{(g(f))^2}$$

5) Calculate Δp_2 = known value − Δp_1

$\mu \sqrt{2L\rho}$] $\sqrt{f}$ new $(g(f))^2$

6) Solve for "LHS" (given Δp_2) → gives f_2 → $\langle v \rangle$ (using friction factor)

7) Solve for Q_{calc} $[Q_{calc} = \langle v \rangle \pi R^2]$

8] Compare this to step 1 Q_{guess} → if close then stop
↳ else iterate

$$\Rightarrow \underbrace{4 \log \left[\sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}} \right] - 0.4}_{\text{"LHS"} \neq f(\langle v_2 \rangle)} = \frac{1}{\sqrt{f}}$$

$\rightarrow f_{new} \approx f$

Rough pipe

Turbulent flow in connected smooth pipes

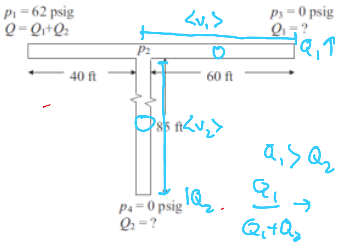
$$\Delta p_2 = \underbrace{p_2 - p_3}_{} = \underbrace{p_2 - p_4}_{}$$

1) Guess Δp_2 "LHS"
↓
find f — friction factor (f, $\Delta p \rightarrow \langle v \rangle$)
↓
$\langle v \rangle$
↓
Q_1 Q_2 — $Q_1 + Q_2 = Q$ → Algorithm 1

2) Total Q ($Q_1 + Q_2 = Q$)
↓
Now use Algorithm 1 !
(until step 5)
↳ $\Delta p_1 = p_1 - p_2$

3) $(\Delta p_2)_{calc} = 62 - \Delta p_1$

4) Iterate until $(\Delta p_2)_{calc} \approx (\Delta p_2)_{guess}$



EXAMPLE 7.9. *A 40-foot section of 1.0-inch ID piping branches into two pipes of the same diameter, one of which is 60.0 feet long and one of which is 85.0 feet long (Figure 7.23). The main pipe is connected to the municipal water supply, which supplies a constant water pressure of 62 psig at the pipe entrance. What are the flow rates through the two pipe exits? What is the pressure at the splitting point? Assume smooth pipes.*

① Since we don't know Q
↳ we first need to solve for Q_1 & Q_2 by guessing Δp_2 value

② Once we know Q → then focus on small initial 40 ft pipe and calculate Δp_2 value independtly
↓
See if it matches?
↳ if not we iterate!

Turbulent flow in connected rough pipes

Colebrook correlation:

$$\frac{1}{\sqrt{f}} = \underbrace{-4 \log \left(\frac{\varepsilon}{D} + \frac{4.67}{Re\sqrt{f}} \right) + 2.28}_{g(f)}$$

PROBLEM: 7.16

For steady pipe flow, repeat the branched-piping example, Example 7.9, for pipes with roughness $\varepsilon = 0.005$.

↓

$$f = \frac{\Delta p D}{2L\rho \langle v \rangle^2} \; ; \; Re = \frac{\rho \langle v \rangle D}{\mu}$$

↓

$$\frac{1}{\sqrt{f}} = \underbrace{-4 \log \left(\frac{\varepsilon}{D} + \frac{4.67}{\left(\sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}} \right)} \right) + 2.28}_{}$$

"LHS" (when we only know pressure drop & no average velocity)